them as the biggest blunder of his life.12 His initial unwillingness to accept the conclusion notwithstanding, Einstein's theory predicted the expansion of the universe. In fact, in the early 1920s—years before Hubble's measurements—the Russian meteorologist Alexander Friedmann had used Einstein's original equations to show, in some detail, that all galaxies would be carried along on the substrate of stretching spatial fabric, thereby speedily moving away from all others. Hubble's observations and numerous subsequent ones have thoroughly verified this astonishing conclusion of general relativity. By offering the explanation for the expansion of the universe, Einstein achieved one of the greatest intellectual feats of all time.

If the fabric of space is stretching, thereby increasing the distance between galaxies that are carried along on the cosmic flow, we can imagine running the evolution backward in time to learn about the origin of the universe. In reverse, the fabric of space shrinks, bringing all galaxies closer and closer to each other. Like the contents of a pressure cooker, as the shrinking universe compresses the galaxies together, the temperature dramatically increases, stars disintegrate and a hot plasma of matter's elementary constituents is formed. As the fabric continues to shrink, the temperature rises unabated, as does the density of the primordial plasma. As we imagine running the clock backward from the age of the presently observed universe, about 15 billion years, the universe as we know it is crushed to an ever smaller size. The matter making up everything—every car, house, building, mountain on earth; the earth itself; the moon; Saturn, Jupiter, and every other planet; the sun and every other star in the Milky Way; the Andromeda galaxy with its 100 billion stars and each and every other of the more than 100 billion galaxies—is squeezed by a cosmic vise to astounding density. And as the clock is turned back to ever earlier times, the whole of the cosmos is compressed to the size of an orange, a lemon, a pea, a grain of sand, and to yet tinier size still. Extrapolating all the way back to "the beginning," the universe would appear to have begun as a point—an image we will critically re-examine in later chapters—in which all matter and energy is squeezed together to unimaginable density and temperature. It is believed that a cosmic fireball, the big bang, erupted from this volatile mixture spewing forth the seeds from which the universe as we know it evolved.